

Skill - 6:

```
sahasrasagi@Sahasra: ~/OS_L  ×  +  ▾
GNU nano 8.7.1 skill_6.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>
#include <stdlib.h>

int main() {
    char command[100];

    printf("Enter command: ");
    fgets(command, sizeof(command), stdin);

    command[strcspn(command, "\n")] = '\0';

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        char *args[20];
        int i = 0;

        char *token = strtok(command, " ");

        while (token != NULL && i < 19) {
            args[i++] = token;
            token = strtok(NULL, " ");
        }

        args[i] = NULL;

        execvp(args[0], args);

        perror("execvp failed");
        exit(1);
    } else {
        wait(NULL);
        printf("Command execution completed.\n");
    }

    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

```
sahasrasagi@Sahasra:~/OS_Lab/skill$ ./skill_6
Enter command: echo 'hello world'
'hello world'
Command execution completed.
sahasrasagi@Sahasra:~/OS_Lab/skill$ |
```

Skill - 7:

```
sahasrasagi@Sahasra: ~/OS_L  ×  +  ∨
GNU nano 8.7.1                                                                    skill
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
int main() {
    char *path = getenv("PATH");

    if (path == NULL) {
        printf("PATHvariable not found.\n");
        return 1;
    }

    printf("PATH Variable:\n%s\n\n", path);

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        printf("Child Process Started\n");
        printf("Child PID: %d\n", getpid());
        // Execute ls
        execlp("ls", "ls", NULL);

        // Runs only if exec fails
        perror("exec failed");
        exit(1);
    } else {
        // Parent process
        int status;

        printf("Parent waiting for child...\n");
        waitpid(pid, &status, 0);

        if (WIFEXITED(status)) {
            printf("Child exited with status: %d\n", WEXITSTATUS(status));
        }
    }

    return 0;
}
```

```

sahasrasagi@Sahasra:~/OS_Lab/skill$ nano skill_7.c
sahasrasagi@Sahasra:~/OS_Lab/skill$ gcc skill_7.c -o skill_7
sahasrasagi@Sahasra:~/OS_Lab/skill$ ./skill_7
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl
ubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files (x86)/VMware/VMware Workstation/bi
les/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path:/mnt/c/Progra
2:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0/;/mnt/c
on/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/mnt/c/Program Files/dotnet/
ogram Files/MySQL/MySQL Shell 8.0/bin/;/mnt/c/Users/sahas/AppData/Local/Programs/Python/Python313/Se
t/c/Users/sahas/AppData/Local/Programs/Python/Launcher/;/mnt/c/Users/sahas/AppData/Local/Microsoft/V
/bin:/mnt/c/Users/sahas/AppData/Roaming/npm:/mnt/c/Users/sahas/AppData/Local/Python/bin:/mnt/c/Progr
/Kingsoft/WPS Office/12.1.0.27458/clitool:/mnt/c/apache-maven-3.9.16-bin/apache-maven-3.9.16/bin:/m

Parent waiting for child...
Child Process Started
Child PID: 665
program1    program2    shellforge    skill_3    skill_4    skill_5    skill_6    skill_7    temp
program1.c  program2.c  shellforge.c  skill_3.c  skill_4.c  skill_5.c  skill_6.c  skill_7.c
Child exited with status: 0
sahasrasagi@Sahasra:~/OS_Lab/skill$ |

```

